

Student Performance Prediction Using Linear Regression

1 Introduction

This project investigates the relationship between students' study hours and their exam scores using Linear Regression. The objective is to build a predictive model capable of estimating academic performance based on study time.

2 Dataset

The dataset contains 96 observations and two variables:

- **Hours:** Number of hours studied.
- **Scores:** Exam score obtained by the student.

A preliminary analysis revealed a positive relationship between study hours and exam scores.

3 Methodology

The dataset was divided into training and testing subsets using an 80/20 split.

A Linear Regression model was trained using Scikit-Learn. The model assumes a linear relationship between the variables:

$$\hat{y} = ax + b \tag{1}$$

where $\hat{y}$ represents the predicted score, a is the slope, and b is the intercept.

4 Results

The fitted regression line showed a strong positive association between study hours and exam scores. Students who studied more generally achieved higher grades.

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

The obtained R^2 score indicated that the model successfully explained a large proportion of the variability in exam scores.

5 Conclusion

This project demonstrates the application of Linear Regression to educational data. The results confirm that study hours are a significant predictor of academic performance. Future work may include additional variables such as attendance, previous grades, and sleep quality to improve prediction accuracy.